

Bessel Functions

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- 2 Bessel Functions of the First and Second Kind
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Introduction

The Bessel functions arise naturally in the solution of problems possessing cylindrical symmetry, such as wave propagation in circular waveguides, vibration of circular membranes, and diffraction by circular apertures. They are defined as the solutions of Bessel's differential equation:

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0, \quad (\text{B.1.1})$$

where ν is the order of the function. The two linearly independent solutions are known as the Bessel function of the first kind $J_\nu(x)$ and the Bessel function of the second kind (or Neumann function) $Y_\nu(x)$.

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Bessel Functions of the First and Second Kind

The Bessel function of the first kind is defined by the series:

$$J_\nu(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(n + \nu + 1)} \left(\frac{x}{2}\right)^{2n+\nu}, \quad (\text{B.2.1})$$

where $\Gamma(z)$ is the Gamma function, defined for $\Re(z) > 0$ by

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt,$$

which generalizes the factorial function such that $\Gamma(n+1) = n!$ for integer $n \geq 0$. For nonnegative real orders ν , $J_\nu(x)$ is finite as $x \rightarrow 0^+$ and therefore commonly appears in physical problems where the field must remain bounded, such as at the center of a circular waveguide or cavity.

The Bessel function of the second kind, or Neumann function, is defined by

$$Y_\nu(x) = \frac{J_\nu(x) \cos(\nu\pi) - J_{-\nu}(x)}{\sin(\nu\pi)}, \quad (\text{B.2.2})$$

Bessel Functions of the First and Second Kind (cont.)

which becomes linearly independent from $J_\nu(x)$ for non-integer ν . For integer order m , it is obtained by the limiting process

$$Y_m(x) = \lim_{\nu \rightarrow m} Y_\nu(x). \quad (\text{B.2.3})$$

The general solution to (B.1.1) is therefore

$$y = c_1 J_\nu(x) + c_2 Y_\nu(x). \quad (\text{B.2.4})$$

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Small- and Large-Argument Behavior

For small arguments ($x \ll 1$), $J_\nu(x)$ behaves as a power series dominated by its leading term:

$$J_\nu(x) \approx \frac{1}{\Gamma(\nu+1)} \left(\frac{x}{2}\right)^\nu, \quad \nu > 0, \quad (\text{B.3.1})$$

which shows that $J_\nu(x) \rightarrow 0$ as $x \rightarrow 0$. In contrast, the Neumann function diverges at the origin:

$$Y_\nu(x) \approx -\frac{\Gamma(\nu)}{\pi} \left(\frac{2}{x}\right)^\nu, \quad \nu > 0. \quad (\text{B.3.2})$$

This singular behavior makes $Y_\nu(x)$ unsuitable for describing fields at $r = 0$, but it is essential in constructing outward- and inward-propagating wave solutions.

For large arguments ($x \gg 1$), both $J_\nu(x)$ and $Y_\nu(x)$ become oscillatory with decreasing amplitude:

$$J_\nu(x) \approx \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{\nu\pi}{2} - \frac{\pi}{4}\right), \quad (\text{B.3.3})$$

$$Y_\nu(x) \approx \sqrt{\frac{2}{\pi x}} \sin\left(x - \frac{\nu\pi}{2} - \frac{\pi}{4}\right). \quad (\text{B.3.4})$$

Small- and Large-Argument Behavior (cont.)

In electromagnetic or acoustic radiation problems, these expressions describe cylindrical waves whose amplitude decays as $1/\sqrt{x}$ with distance — a slower decay than the $1/x$ dependence of spherical waves.

The Hankel functions are defined as

$$H_\nu^{(1,2)}(x) = J_\nu(x) \pm iY_\nu(x). \quad (\text{B.3.5})$$

With the $e^{i\omega t}$ convention, they represent inward and outward cylindrical waves, respectively.

For small arguments ($x \ll 1$), they behave as

$$H_\nu^{(1,2)}(x) \approx \mp i \frac{\Gamma(\nu)}{\pi} \left(\frac{2}{x}\right)^\nu + \frac{1}{\Gamma(\nu+1)} \left(\frac{x}{2}\right)^\nu, \quad \nu > 0, \quad (\text{B.3.6})$$

showing a singular behavior near the origin dominated by the Neumann-type term. Their asymptotic forms for large x are

$$H_\nu^{(1,2)}(x) \approx \sqrt{\frac{2}{\pi x}} e^{\pm i\left(x - \frac{\nu\pi}{2} - \frac{\pi}{4}\right)}, \quad (\text{B.3.7})$$

When the $e^{i\omega t}$ convention is adopted, the exponential phase terms clearly distinguish their physical roles:

Small- and Large-Argument Behavior (cont.)

- $H_\nu^{(1)}(x)$ represents an *inward-propagating* cylindrical wave, corresponding to convergence toward the source.
- $H_\nu^{(2)}(x)$ represents an *outward-propagating* cylindrical wave, corresponding to radiation away from the source.

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Summary of Important Relations

1 Recurrence relations:

$$\begin{aligned}J_{\nu-1}(x) + J_{\nu+1}(x) &= \frac{2\nu}{x} J_{\nu}(x), \\J_{\nu-1}(x) - J_{\nu+1}(x) &= 2J'_{\nu}(x).\end{aligned}$$

Similar relations hold for $Y_{\nu}(x)$, $H_{\nu}^{(1)}(x)$, and $H_{\nu}^{(2)}(x)$.

2 Wronskian relations:

$$\begin{aligned}J_{\nu}(x)Y'_{\nu}(x) - J'_{\nu}(x)Y_{\nu}(x) &= \frac{2}{\pi x}, \\J_{\nu}(x)H_{\nu}^{(1,2)'}(x) - J'_{\nu}(x)H_{\nu}^{(1,2)}(x) &= \pm \frac{2i}{\pi x}.\end{aligned}$$

3 Generating function:

$$e^{\frac{x}{2}(t-1/t)} = \sum_{n=-\infty}^{\infty} J_n(x)t^n.$$

Summary of Important Relations (cont.)

4 Integral representation (integer n):

$$J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x \sin \theta) d\theta,$$

$$\begin{aligned} J_n(x) &= \frac{1}{2\pi i^n} \int_0^{2\pi} e^{i(x \cos \theta + n\theta)} d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} e^{i(x \sin \theta - n\theta)} d\theta. \end{aligned}$$

5 Derivative relation:

$$\frac{d}{dx} [x^\nu J_\nu(x)] = x^\nu J_{\nu-1}(x).$$

6 Orthogonality:

$$\int_0^a J_\nu\left(\frac{\alpha_{\nu m}\rho}{a}\right) J_\nu\left(\frac{\alpha_{\nu n}\rho}{a}\right) \rho d\rho = \frac{a^2}{2} [J_{\nu+1}(\alpha_{\nu n})]^2 \delta_{mn},$$

where $\alpha_{\nu n}$ is the n -th zero of J_ν .

Summary of Important Relations

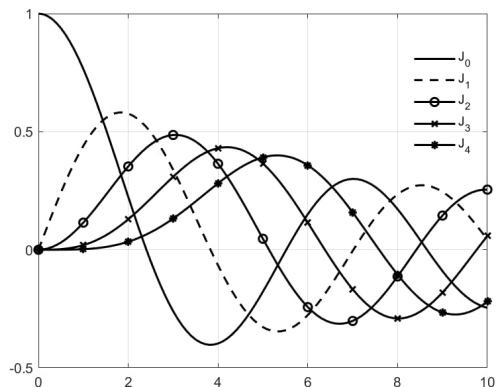


Figure B.1: Bessel functions of the first kind for $\nu = 0, 1, 2, 3, 4$.

Summary of Important Relations

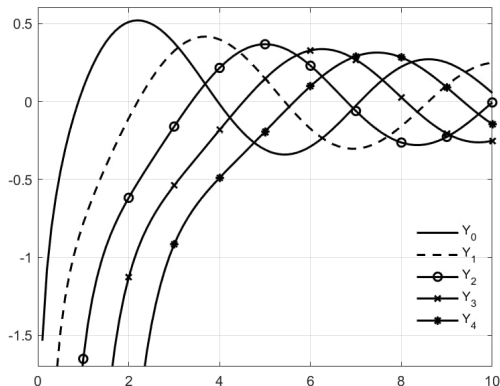


Figure B.2: Bessel functions of the second kind for $\nu = 0, 1, 2, 3, 4$.

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